Tasks:

1. **Open Burp Suite.**
2. **Navigate to Decoder.**
3. **Perform the following operations:**
- **URL Encode**
- **URL Decode**
- **Base64 Encode**
- **Base64 Decode**
- **HTML Encode**
- **HTML Decode**
4. **Take screenshots of each conversion.**
5. **Create 5 sample strings and perform all encoding and decoding operations.**

Examples:

- **CyberSecurity**
- **VExTechSolution**
- **Password123**
- **Student01**
- **HelloWorld**

6. **Navigate to Comparer.**
7. **Compare:**
- **Two different login requests**
- **Two different responses**
- **Two different encoded values**
8. **Observe and note:**
- **Similarities**
- **Differences**
- **Modified parameters**

Assignment:

Prepare a report containing:

1. **What is Encoding?**
2. **Difference between Encoding, Encryption and Hashing.**
3. **What is Base64?**
4. **What is URL Encoding?**
5. **Why do web applications use encoding?**
6. **Screenshots of Decoder operations.**
7. **Screenshots of Comparer results.**

Using Burp Decoder, decode the following values and explain the output:

- **SGVsbG8gV29ybGQ=**
- **Q3liZXIgU2VjdXJpdHk=**
- **VkVYIFRlY2ggU29sdXRpb24=**
- **UGFzc3dvcmQxMjM=**

Aim

To understand the use of Burp Suite Decoder and Comparer tools by performing encoding/decoding operations and comparing different HTTP messages.

Objective

- To learn various encoding and decoding techniques using Burp Decoder.
- To understand the working of Burp Comparer.
- To identify similarities and differences between HTTP requests and responses.
- To understand the concepts of Encoding, Encryption, and Hashing.

What is Encoding?

Encoding is the process of converting data from one format into another using a publicly available algorithm so that systems can correctly store, transmit, and interpret the information.

Encoding does not provide security. It only changes the representation of data.

Examples:

- Base64 Encoding
- URL Encoding
- HTML Encoding

Difference Between Encoding, Encryption and Hashing

Feature	Encoding	Encryption	Hashing
Purpose	Data representation	Data confidentiality	Data integrity
Reversible	Yes	Yes (with key)	No
Uses Key	No	Yes	No
Security	No	Yes	One-way verification
Example	Base64	AES, RSA	SHA-256, MD5

What is Base64?

Base64 is an encoding technique that converts binary data into ASCII text format.

It is commonly used for:

- Email attachments
- API tokens
- Basic Authentication
- Data transmission

Example:

Text: Hello World

Base64:
SGVsbG8gV29ybGQ=

What is URL Encoding?

URL Encoding converts special characters into a format that can be safely transmitted through URLs.

Format:
% followed by hexadecimal value.

Example:

Space = %20

@ = %40

? = %3F

Why Do Web Applications Use Encoding?

Web applications use encoding to:

- Safely transmit data over networks.
- Handle special characters.
- Prevent misinterpretation of user input.
- Preserve data during communication.
- Maintain compatibility among systems.

Burp Decoder Operations

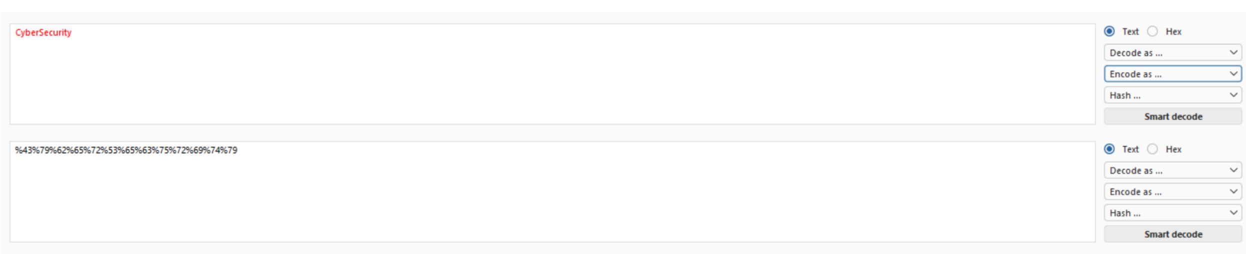
Procedure

1. Open Burp Suite Community Edition.
2. Navigate to Decoder.
3. Enter a sample string.
4. Select the required conversion.
5. Observe the output.
6. Take screenshots for each operation.

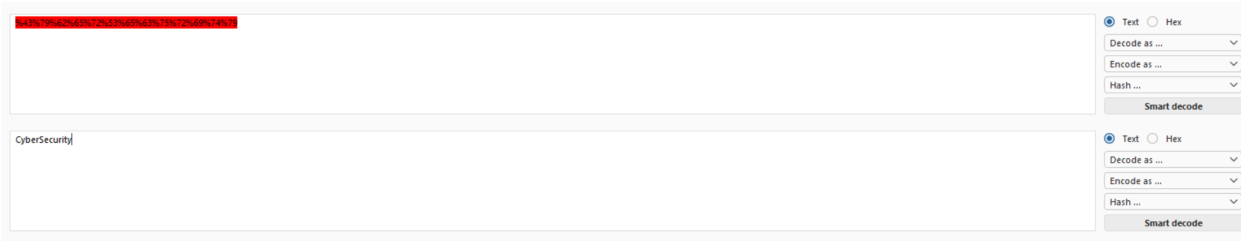
Decoder Results

Sample 1: CyberSecurity

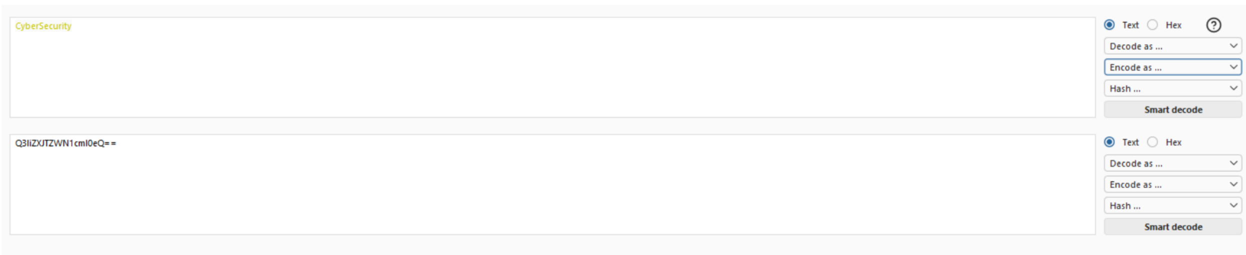
URL Encode:



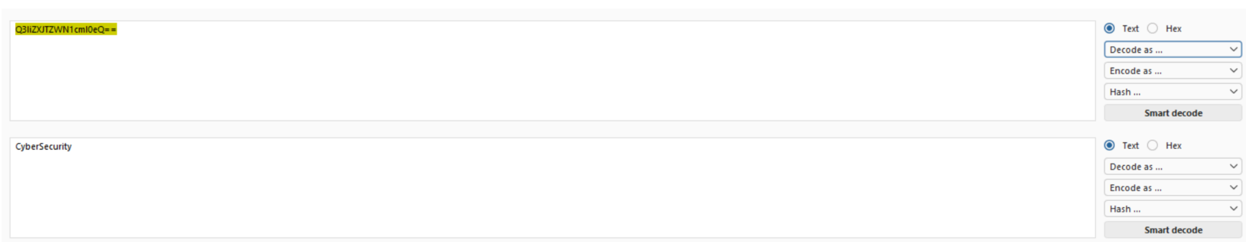
URL Decode:



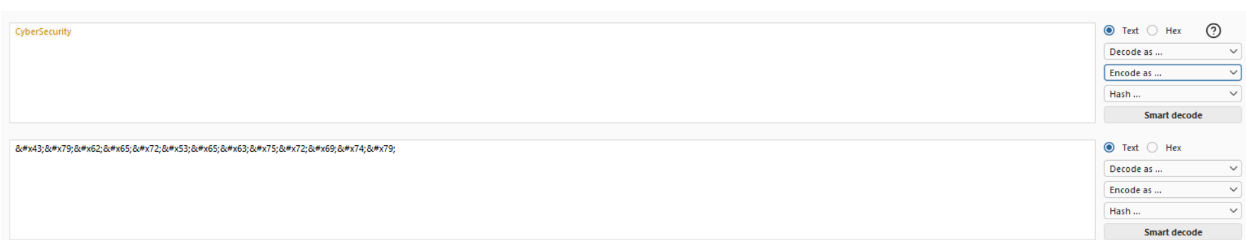
Base64 Encode:



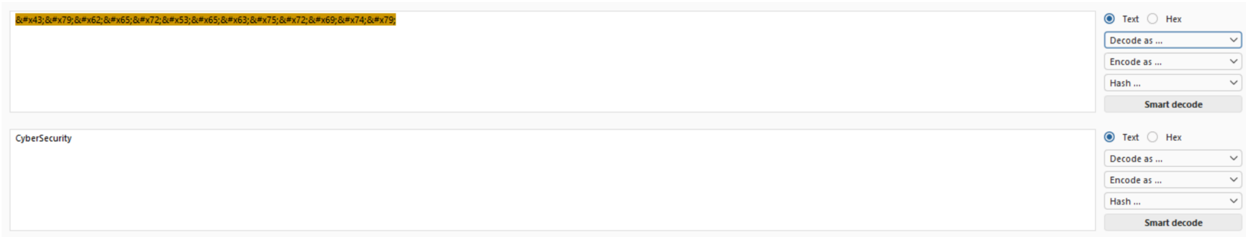
Base64 Decode:



HTML Encode:

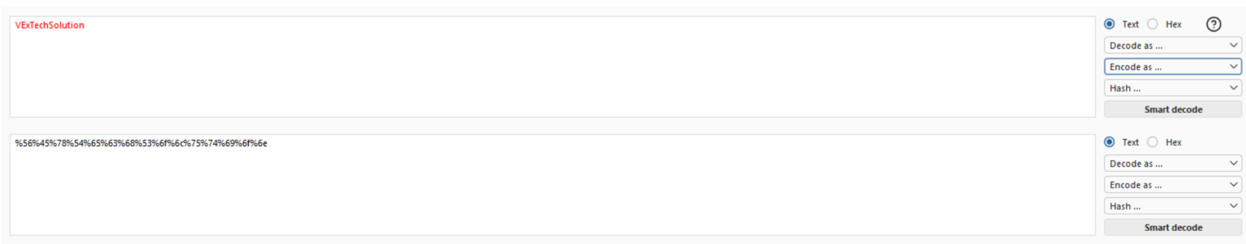


HTML Decode:

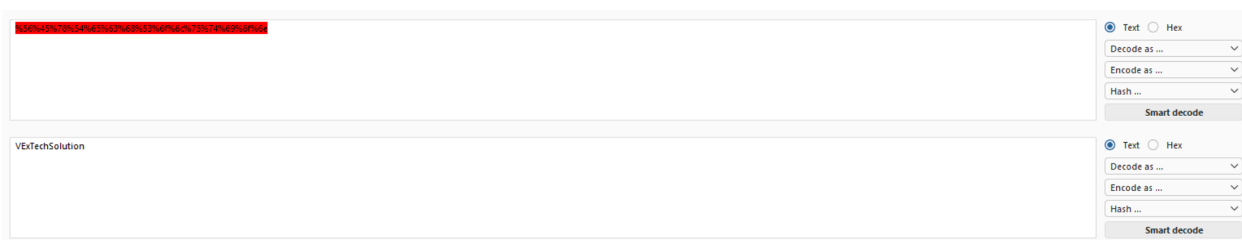


Sample 2: VExTechSolution

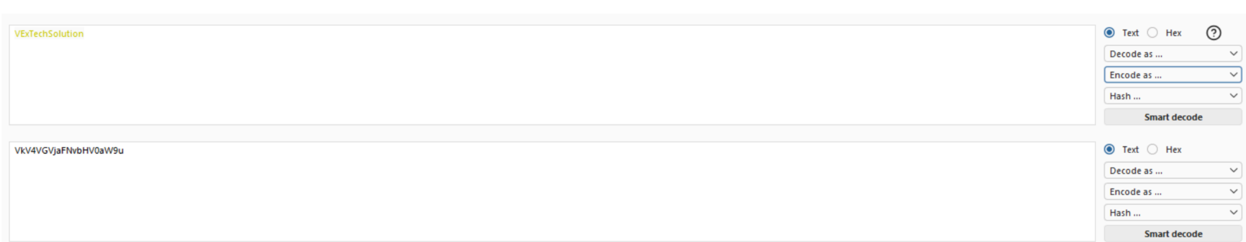
URL Encode:



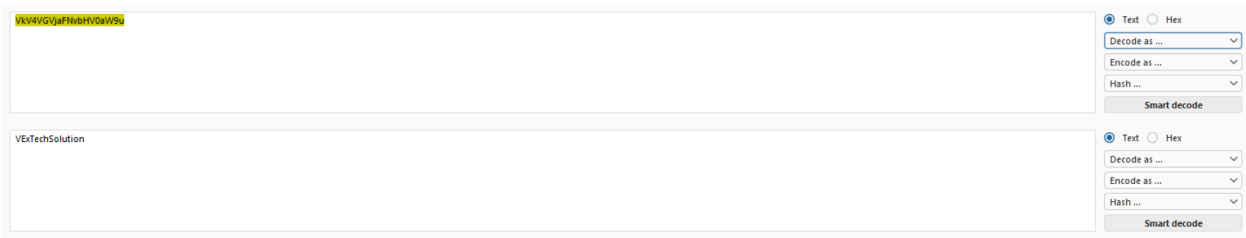
URL Decode:



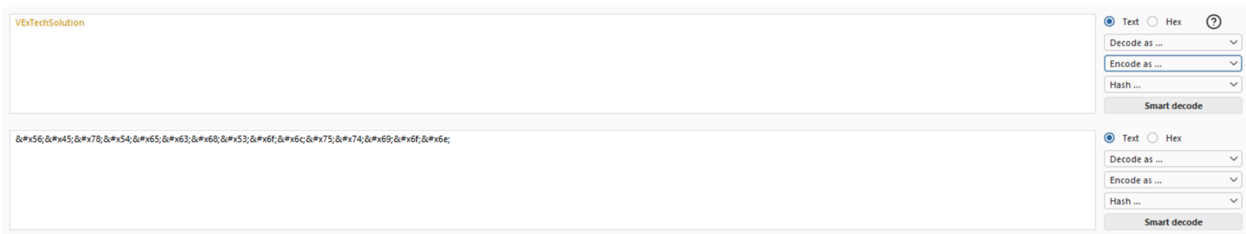
Base64 Encode:



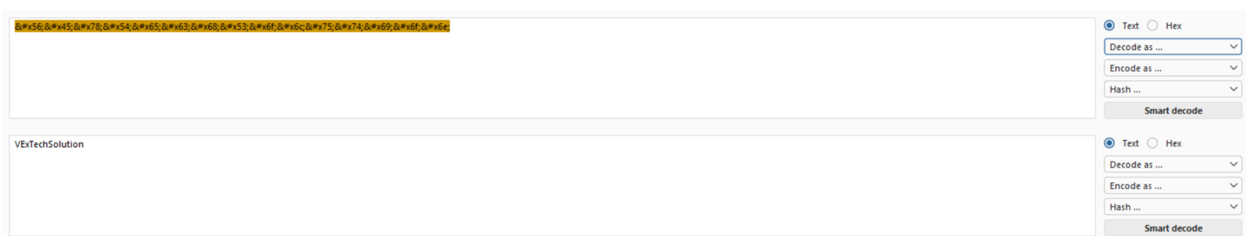
Base64 Decode:



HTML Encode:

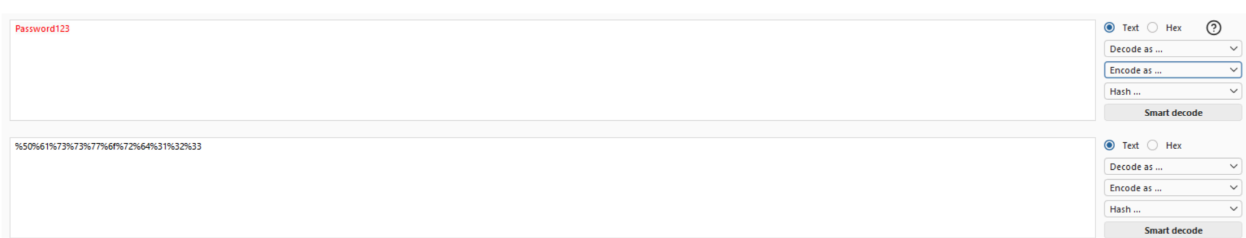


HTML Decode:

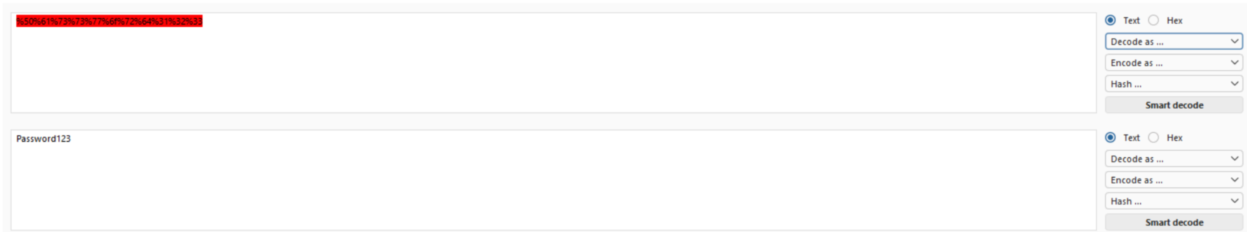


Sample 3: Password123

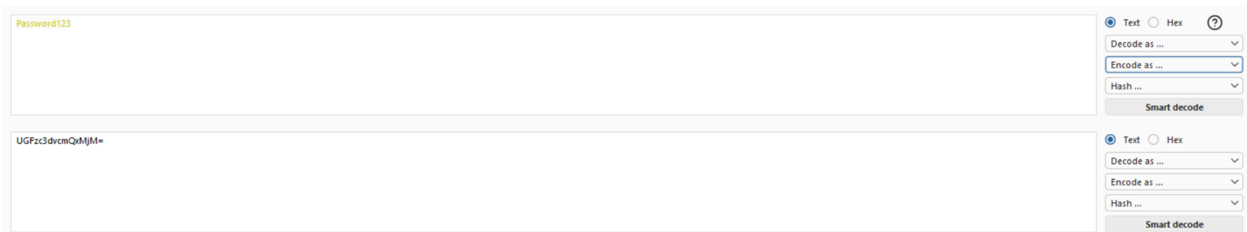
URL Encode:



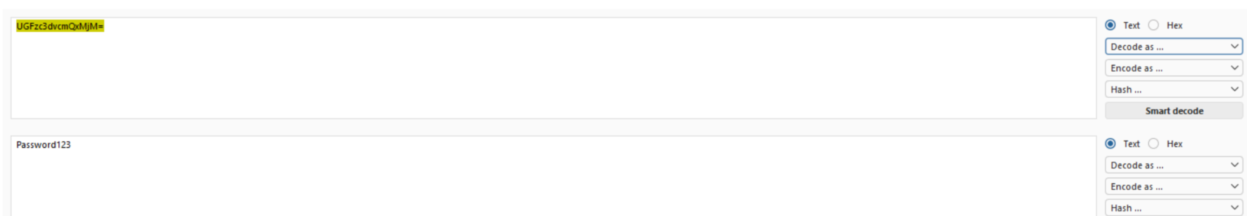
URL Decode:



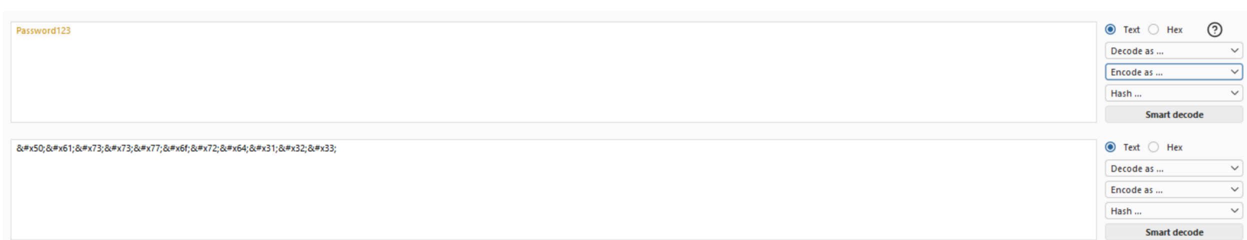
Base64 Encode:



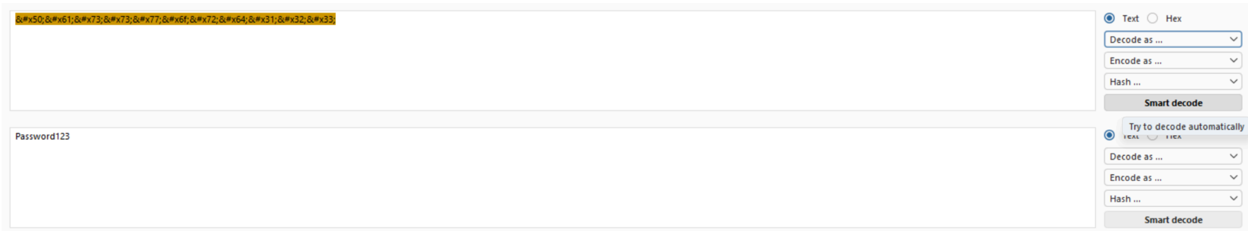
Base64 Decode:



HTML Encode:

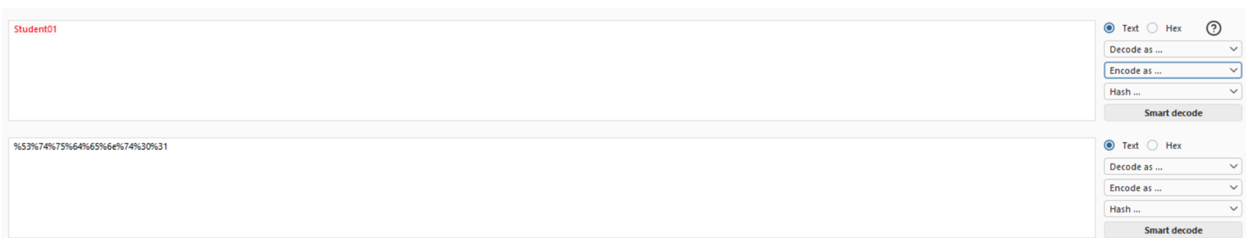


HTML Decode:

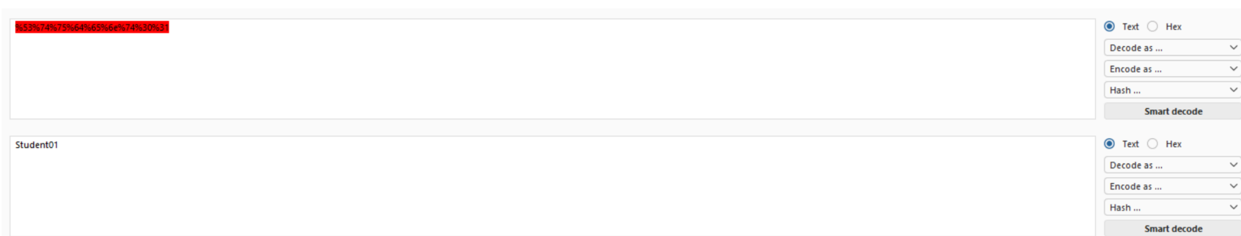


Sample 4: Student01

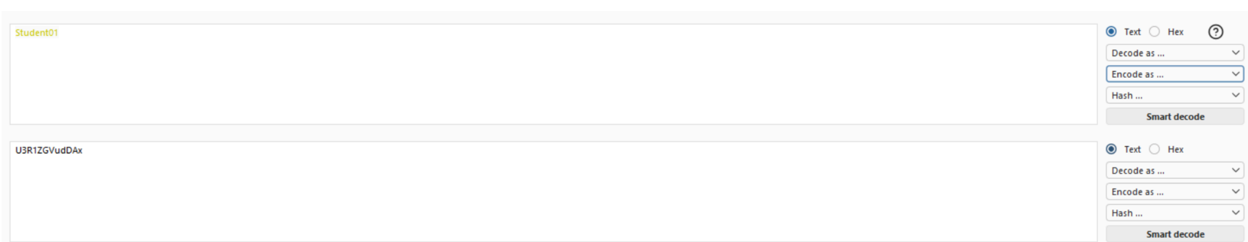
URL Encode:



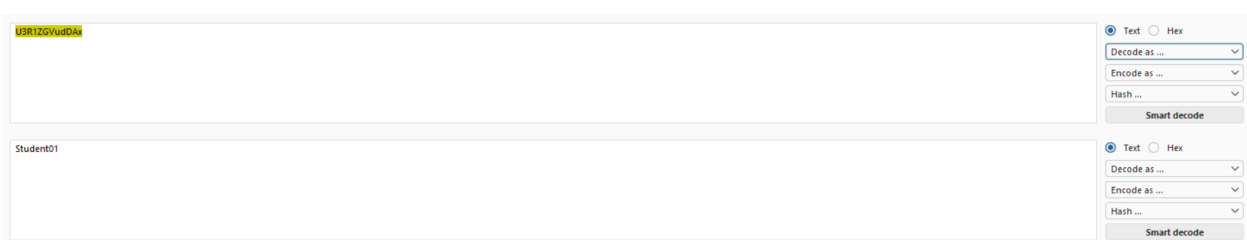
URL Decode:



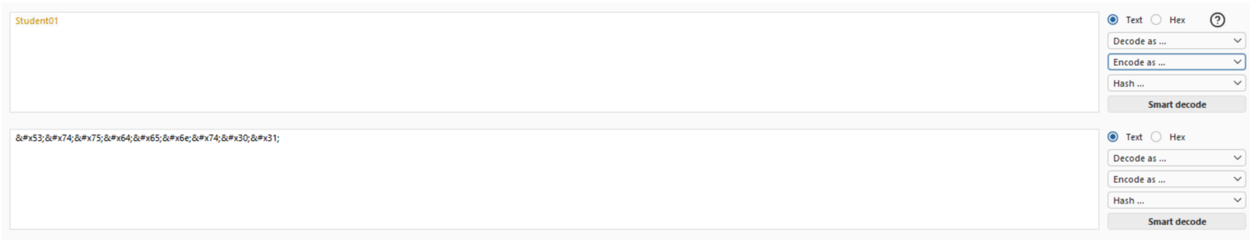
Base64 Encode:



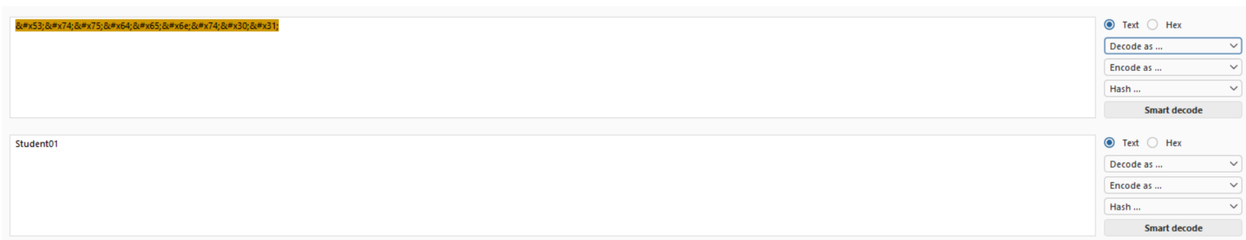
Base64 Decode:



HTML Encode:

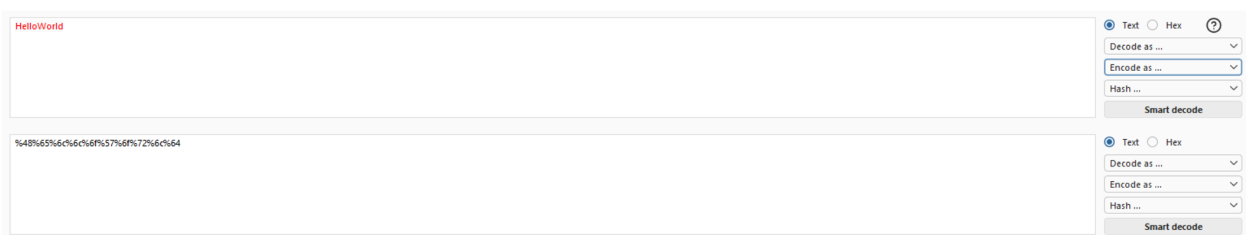


HTML Decode:

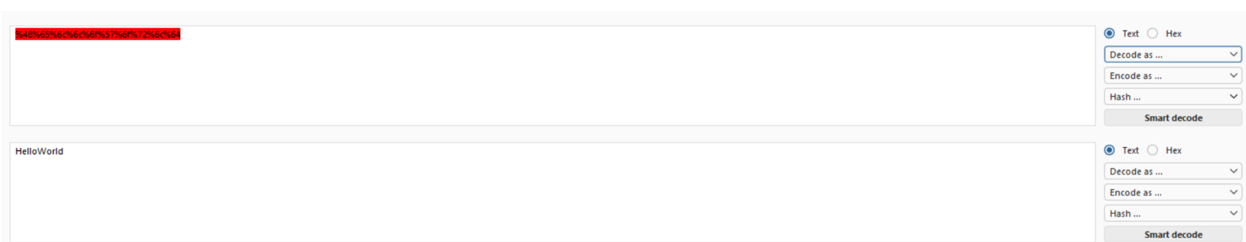


Sample 5: HelloWorld

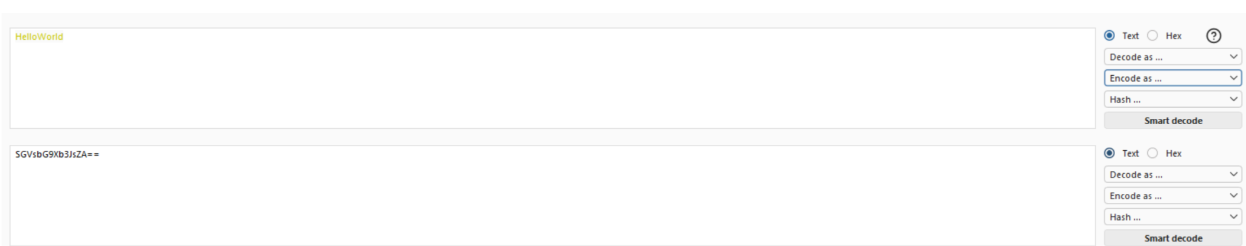
URL Encode:



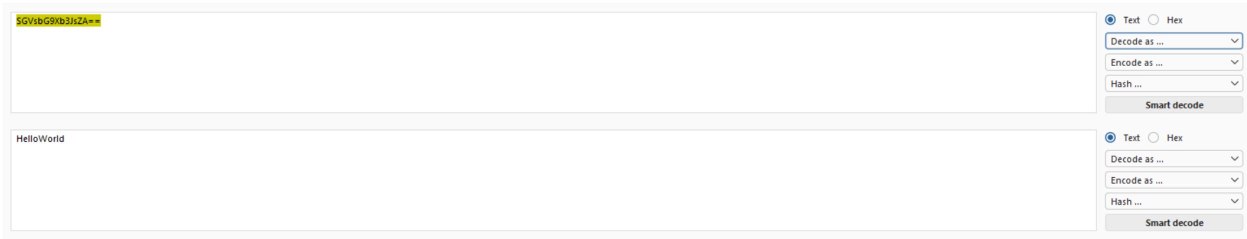
URL Decode:



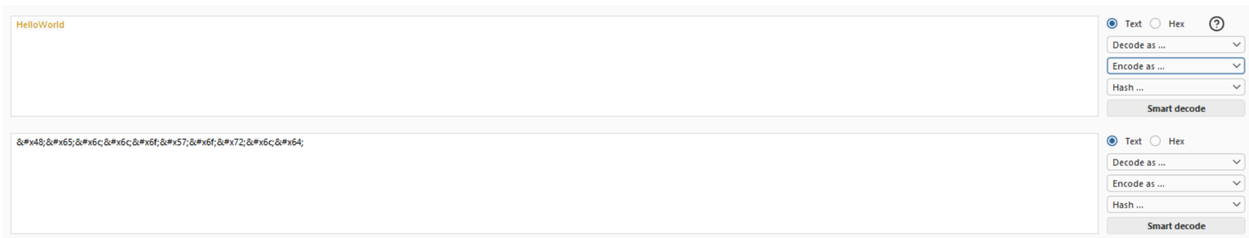
Base64 Encode:



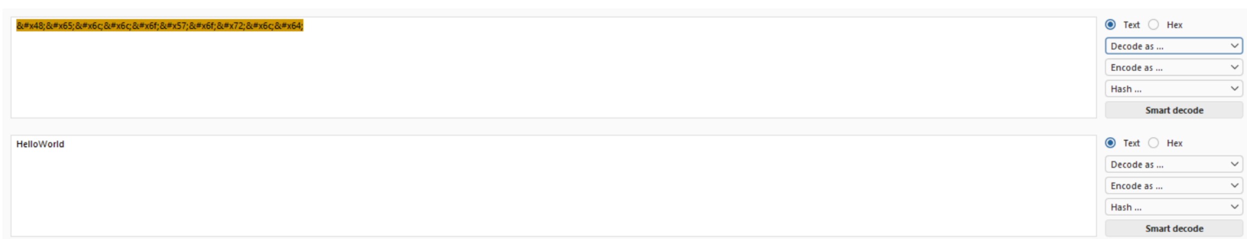
Base64 Decode:



HTML Encode:



HTML Decode:



Burp Comparer

Procedure

1. Open Burp Suite.
2. Navigate to Comparer.
3. Send items to Comparer.
4. Select "Words" comparison.
5. Observe highlighted differences.

Comparison 1: Login Requests

Create First Request

Click: Paste

Paste: Request 1

Click **OK**.

Create Second Request

Again click: Paste

Paste: Request 2

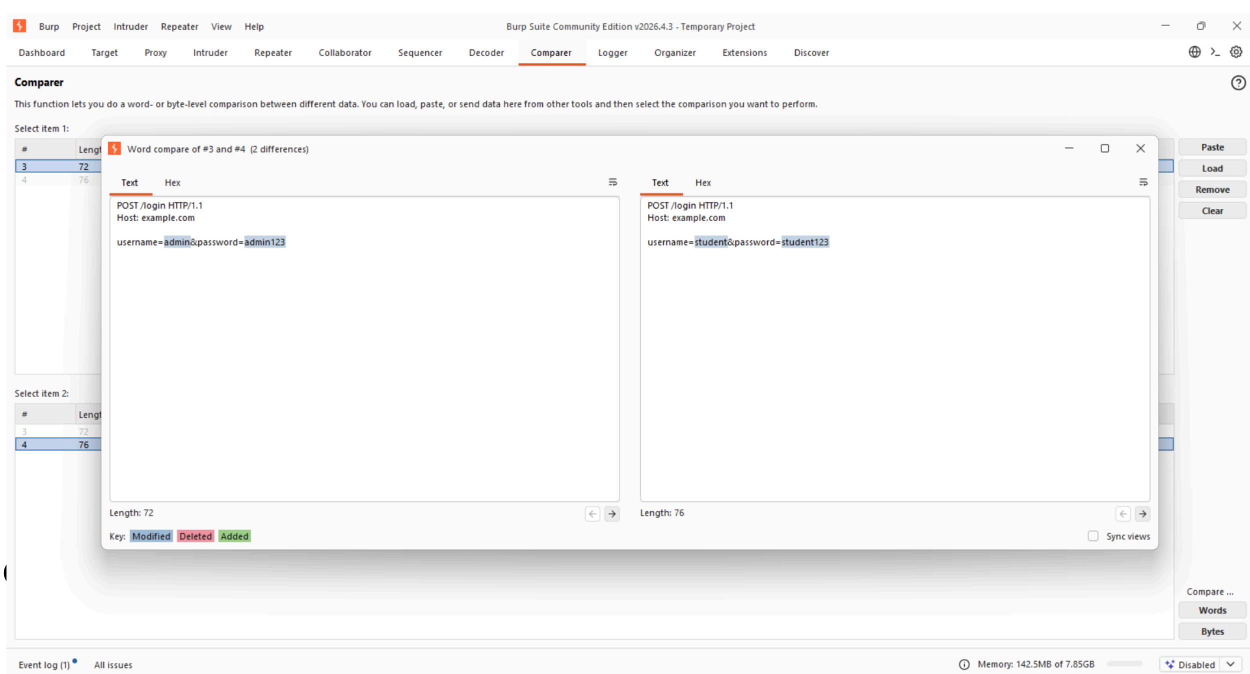
Click **OK**.

Compare

You will now see both items.

Click: **Words**

Burp highlights differences.



Paste: **Response 1**

Click **OK**

Paste: **Response 2**

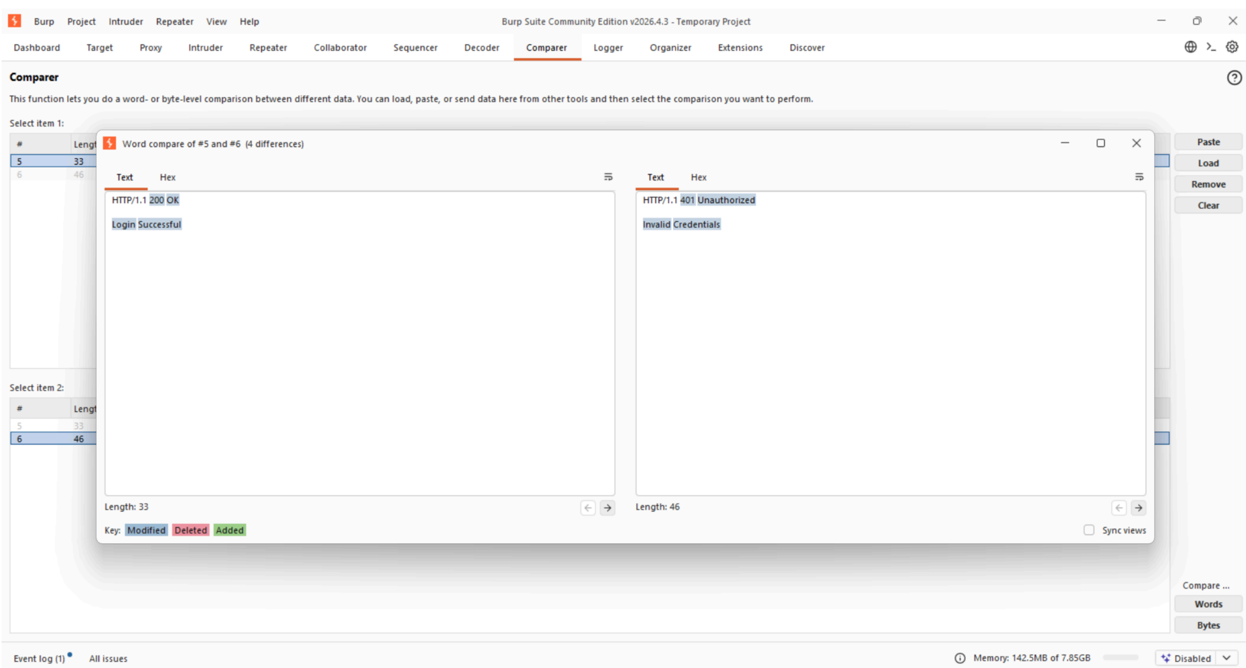
Click **OK**

Compare

You will now see both items.

Click: **Words**

Burp highlights differences.



Comparison 3: Encoded Values

Paste:

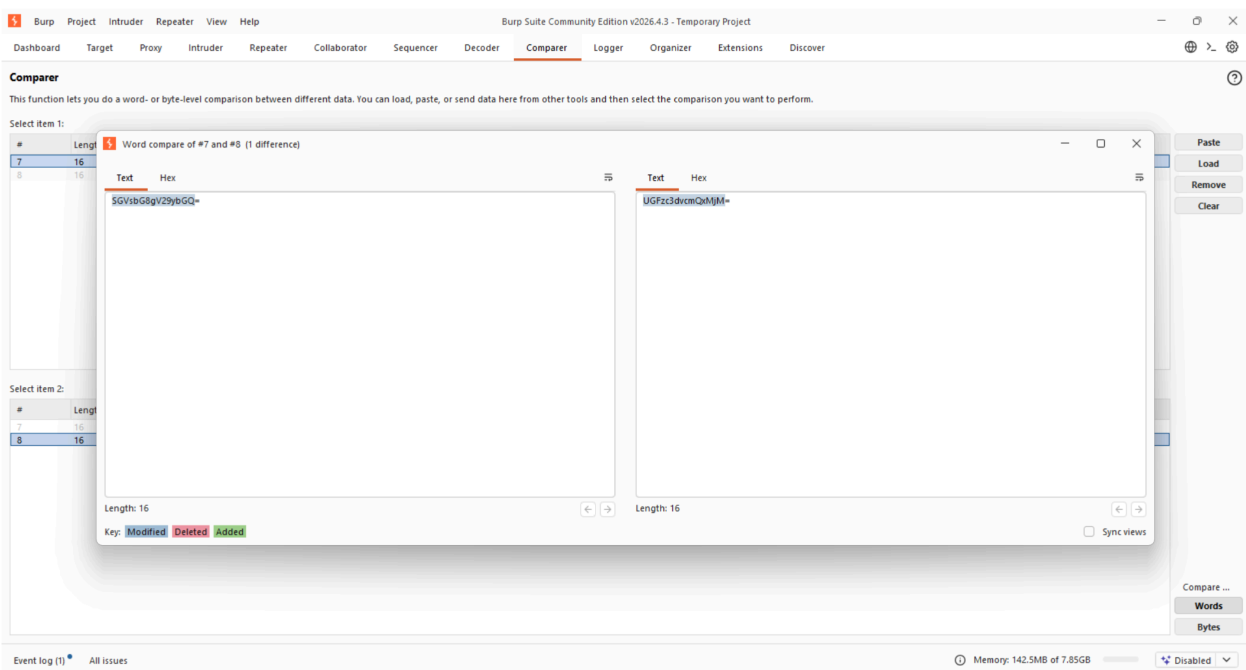
First: **SGVsbG8gV29ybGQ=**

Second: **UGFzc3dvcmQxMjM=**

Select both.

Click: **Words**

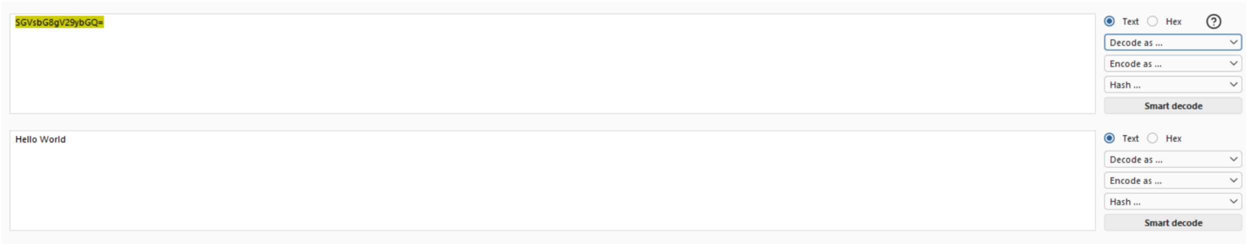
Burp shows the differences.



Decoding Given Values Using Burp Decoder

1. SGVsbG8gV29ybGQ=

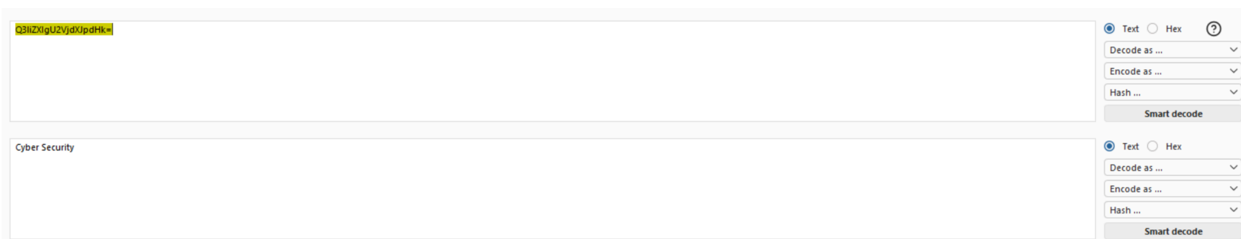
Decoded Output:
Hello World



Explanation:
This is the Base64 encoded representation of "Hello World".

2. Q3liZXIgU2VjdXJpdHk=

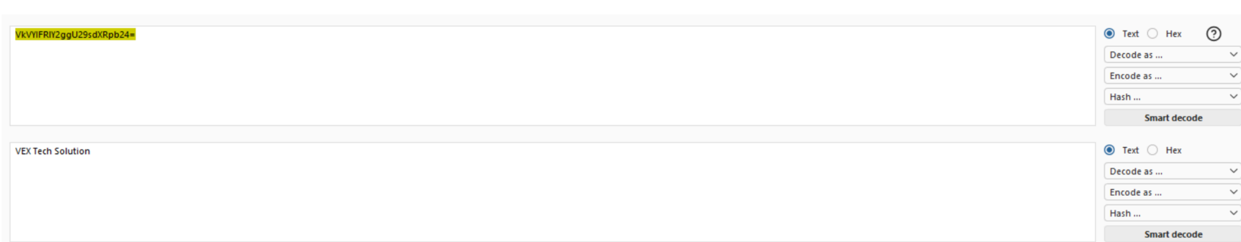
Decoded Output:
Cyber Security



Explanation:
This Base64 value decodes to the phrase "Cyber Security".

3. VkVYIFRlY2ggU29sdXRpb24=

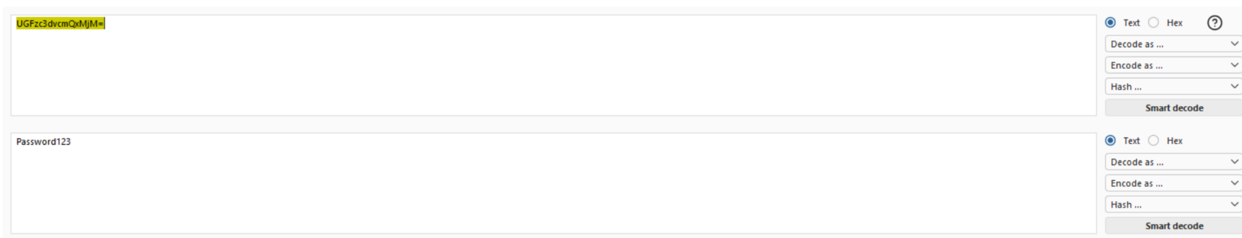
Decoded Output:
VEX Tech Solution



Explanation:
This Base64 string represents the company name "VEX Tech Solution".

4. UGFzc3dvcmQxMjM=

Decoded Output:
Password123



Explanation:
This Base64 value decodes to the password text "Password123".

Conclusion

The practical was successfully completed using Burp Suite Community Edition. The Decoder tool was used to perform URL Encoding/Decoding, Base64 Encoding/Decoding, and HTML Encoding/Decoding on different sample strings. This helped in understanding how data is represented and transmitted in web applications. The Comparer tool was used to compare login requests, responses, and encoded values to identify similarities, differences, and modified parameters.